

SIMPLE LINEAR MODEL VIA ORDINARY LEAST SQUARES (OLS)

Assume that on n statistical units (individuals) we observe (x_i, y_i) , $i=1, \dots, n$.

Hence the data are $\underline{y} = (y_1, \dots, y_n)$ and $\underline{x} = (x_1, \dots, x_n)$.

We consider that each y_i is realization of a r.v. Y_i , $i=1, \dots, n \rightarrow$ sample space $\mathcal{Y} = \mathbb{R}^n$

We do not specify a distribution for $(Y_1, \dots, Y_n)$: we only make assumptions about the first two moments $E[Y_i]$ and $var(Y_i)$.

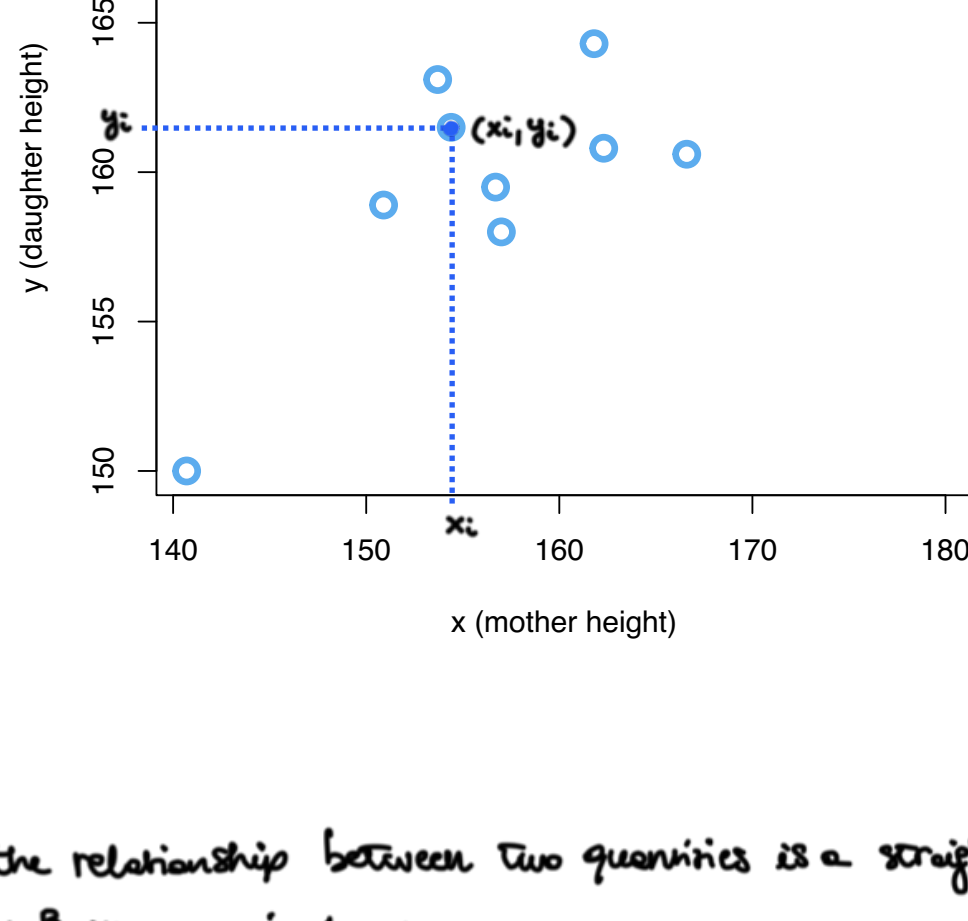
We specify a simple linear model (only 1 covariate)

We estimate the model parameters only through "intuitive" considerations and a simple optimization ("ordinary least squares" method)

We start with a simple example

relationship between the height of 11 mothers (x_i) and the height of their daughters (y_i).

	x	y
1	153.7	163.1
2	156.7	159.5
3	173.5	169.4
4	157.0	158.0
5	161.8	164.3
6	140.7	150.0
7	179.8	170.3
8	150.9	158.9
9	154.4	161.5
10	162.3	160.8
11	166.6	160.6



Intuition:

the simplest way to describe the relationship between two quantities is a straight line:

$$Y_i = \beta_1 + \beta_2 x_i \quad i=1, \dots, n$$

However, such a relationship does not hold exactly: the points are not PERFECTLY ALIGNED.

hence we add an error term to take into account this discrepancy:

$$Y_i = \beta_1 + \beta_2 x_i + \epsilon_i \quad i=1, \dots, n$$

1st step: MODEL SPECIFICATION

Consider the model:

$$Y_i = \beta_1 + \beta_2 x_i + \epsilon_i \quad i=1, \dots, n$$

(β_1, β_2) are the REGRESSION COEFFICIENTS

We specified a straight line with the intercept (β_1)

we only observe 1 covariate, but we also introduce one additional variable taking value 1 for each individual.

The model matrix then is:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$$

$\Rightarrow \beta_1$ is the INTERCEPT (coefficient of 1)

β_2 is the COEFFICIENT of x (slope)

ASSUMPTIONS on the independent variables

- $x_1, \dots, x_n$ fixed and non-stochastic
- the x_i can not be all equal (sample variance of $(x_1, \dots, x_n)$ must be $\neq 0$)

The systematic component is now fully specified, we need to define the stochastic component (ϵ) .

ASSUMPTIONS on the STOCHASTIC COMPONENT

- $E[\epsilon_i] = 0$ for $i=1, \dots, n$
- $Var(\epsilon_i) = \sigma^2 > 0$ $i=1, \dots, n$ (common variance across subjects)
- $Cov(\epsilon_i, \epsilon_k) = 0$ if $i \neq k$, $i=1, \dots, n$ $k=1, \dots, n$

- $E[\epsilon_i] = 0$ $i=1, \dots, n$ ABSENCE OF SYSTEMATIC ERROR

Implications for Y_i

$$E[Y_i] = E[\beta_1 + \beta_2 x_i + \epsilon_i] = E[\underbrace{\beta_1 + \beta_2 x_i}_{\text{non-stochastic}}] + E[\underbrace{\epsilon_i}_{\text{stochastic}}] = \beta_1 + \beta_2 x_i$$

What happens if there is a systematic error? i.e. $E[\epsilon_i] = c \neq 0$

$$E[Y_i] = \beta_1 + \beta_2 x_i + c = (\beta_1 + c) + \beta_2 x_i$$

the systematic error c is incorporated into the intercept (not a problem)

it is equivalent to a model

$$Y_i = \beta_1^* + \beta_2 x_i + \epsilon_i^* \quad \text{where } \beta_1^* = \beta_1 + c$$

$$\epsilon_i^* = \epsilon_i - c \Rightarrow E[\epsilon_i^*] = 0$$

- $Var(\epsilon_i) = \sigma^2 > 0$ for all $i=1, \dots, n$ HOMOSEDASTICITY OF THE ERRORS

Implications for Y_i :

$$var(Y_i) = var(\underbrace{\beta_1 + \beta_2 x_i}_{\text{non-stoch.}} + \epsilon_i) = var(\epsilon_i) = \sigma^2 \quad \forall i=1, \dots, n$$

$\Rightarrow$ homoscedasticity of the response

- $Cov(\epsilon_i, \epsilon_k) = 0$ for $i \neq k$ THE ERRORS ARE UNCORRELATED

Implication for Y_i

$$Cov(Y_i, Y_k) = Cov(\underbrace{\beta_1 + \beta_2 x_i}_{\text{non-stochastic}} + \epsilon_i, \underbrace{\beta_1 + \beta_2 x_k + \epsilon_k}_{\text{non-stochastic}}) = Cov(\epsilon_i, \epsilon_k) = 0$$

2nd step: ESTIMATE

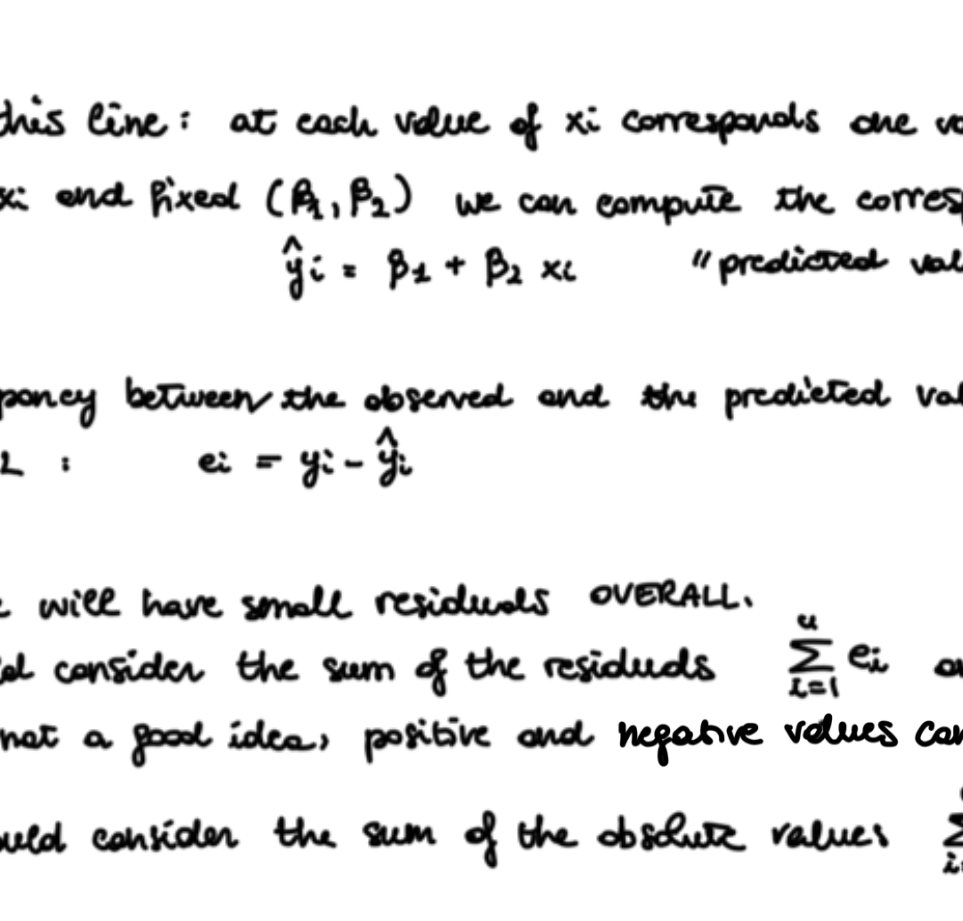
what do we need to estimate? Unknown quantities are $(\beta_1, \beta_2, \sigma^2)$

Hence the PARAMETER SPACE is $\Theta = \mathbb{R}^2 \times (0, +\infty)$

Every combination of (β_1, β_2) determines a specific line: how do we select the "best" line?

We need a criterion of what is a "good" line.

We want a line which is the closest to the observed points.



Consider this line: at each value of x_i corresponds the value of y_i that lies on the line

$\Rightarrow$ given x_i and fixed (β_1, β_2) we can compute the corresponding value of y_i (according to the line)

$$\hat{y}_i = \beta_1 + \beta_2 x_i \quad \text{"predicted value"}$$

The discrepancy between the observed and the predicted value (at the observed locations x_i)

$$\text{RESIDUAL: } \epsilon_i = y_i - \hat{y}_i$$

A good line will have small residuals OVERALL.

- we could consider the sum of the residuals $\sum_{i=1}^n \epsilon_i$ and select the (β_1, β_2) that minimize it

$\rightarrow$ not a good idea, positive and negative values cancel out.

- we could consider the sum of the absolute values $\sum_{i=1}^n |\epsilon_i| \rightarrow$ mathematically not very practical

- we consider instead the sum of the SQUARED residuals

$$\sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2 = S(\beta_1, \beta_2)$$

and take as an estimate of (β_1, β_2) the combination that minimizes it.

DEF: the LEAST SQUARES estimate of (β_1, β_2) is the combination of values $(\hat{\beta}_1, \hat{\beta}_2)$ that minimizes $S(\beta_1, \beta_2)$

$$(\hat{\beta}_1, \hat{\beta}_2) = \underset{(\beta_1, \beta_2) \in \mathbb{R}^2}{\operatorname{argmin}} S(\beta_1, \beta_2)$$

$$= \underset{(\beta_1, \beta_2) \in \mathbb{R}^2}{\operatorname{argmin}} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2$$

We have hence turned a problem of estimation into an optimization.

THEM: The least squares estimate of (β_1, β_2) is

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}$$

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad (\text{sample mean}).$$

Remark:

recall that the sample variance of $(x_1, \dots, x_n)$ is $s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

(and similarly for s_y^2)

the sample covariance is $s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

$$\text{Hence } \hat{\beta}_2 = \frac{s_{xy}}{s_x^2}$$

Proof: we want to show that $\hat{\beta}_1, \hat{\beta}_2$ minimize $S(\beta_1, \beta_2) = \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2$.

We need to find the critical points (1st derivative = 0)

and then check that $(\hat{\beta}_1, \hat{\beta}_2)$ is a minimum (2nd derivative > 0)

$$\begin{cases} \frac{\partial S(\beta_1, \beta_2)}{\partial \beta_1} = 0 \\ \frac{\partial S(\beta_1, \beta_2)}{\partial \beta_2} = 0 \end{cases} \Leftrightarrow \begin{cases} \sum_{i=1}^n 2(y_i - \beta_1 - \beta_2 x_i)(-1) = 0 \\ \sum_{i=1}^n 2(y_i - \beta_1 - \beta_2 x_i)(-x_i) = 0 \end{cases}$$

$$\begin{cases} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i) = 0 & \textcircled{A} \\ \sum_{i=1}^n x_i (y_i - \beta_1 - \beta_2 x_i) = 0 & \textcircled{B} \end{cases}$$

$$\textcircled{A} \quad n\bar{y} - n\beta_1 - \beta_2 \sum_{i=1}^n x_i = 0 \quad (\text{since } \sum_{i=1}^n y_i = n\bar{y})$$

$$\beta_1 = \bar{y} - \beta_2 \bar{x}$$

$$\textcircled{B} \quad \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} - \beta_2 \sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} + n\beta_2 \bar{x}^2 - \beta_2 \sum_{i=1}^n x_i^2 = 0$$

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$$

$$\text{we obtain } \hat{\beta}_2 = \frac{s_{xy}}{s_x^2}$$

$$\text{and } \hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}$$

Is $(\hat{\beta}_1, \hat{\beta}_2)$ a minimum? We compute the Hessian

$$H = \begin{bmatrix} \frac{\partial^2 S(\beta_1, \beta_2)}{\partial \beta_1^2} & \frac{\partial^2 S(\beta_1, \beta_2)}{\partial \beta_1 \partial \beta_2} \\ \frac{\partial^2 S(\beta_1, \beta_2)}{\partial \beta_2 \partial \beta_1} & \frac{\partial^2 S(\beta_1, \beta_2)}{\partial \beta_2^2} \end{bmatrix} = \begin{bmatrix} 2n & 2n\bar{x} \\ 2n\bar{x} & 2 \sum_{i=1}^n x_i^2 \end{bmatrix}$$

$$\det(H) = 4n \sum_{i=1}^n x_i^2 - 4n^2 \bar{x}^2 = 4n \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right) = 4n \sum_{i=1}^n (x_i - \bar{x})^2 > 0$$

since $\det(H) > 0$ and $H_{2,2} = 2 \sum_{i=1}^n x_i^2 > 0$, $(\hat{\beta}_1, \hat{\beta}_2)$ is a minimum of $S(\beta_1, \beta_2)$

Moreover, it is the global minimum.

Remarks:

- we did not use the assumptions on ϵ_i

- we used the assumption on the x_i : what happens if $x_i = x_0$ for all $i=1, \dots, n$?

$$(x_i - \bar{x}) = 0 \quad \forall i \Rightarrow s_x^2 = 0 \quad \text{and } s_{xy} = 0 \Rightarrow \hat{\beta}_2 = \frac{0}{0} \quad \text{not defined}$$

- once we estimate $(\hat{\beta}_1, \hat{\beta}_2)$, we automatically obtain $\hat{y} = \hat{\beta}_1 + \hat{\beta}_2 x$, i.e. the estimated regression line.

- $\hat{y}$ allows us to make predictions: given a generic value x , we predict the corresponding value of the response.

As usual, careful with extrapolation, i.e., estimating the response for a value of x outside of the observed range of $(x_1, \dots, x_n)$.

• INTERPRETATION of $(\hat{\beta}_1, \hat{\beta}_2)$

we have estimated a line $\hat{y} = \hat{\beta}_1 + \hat{\beta}_2 x$

- INTERPRETATION OF $\hat{\beta}_1$

$\hat{\beta}_1$ is the intercept, i.e., the predicted value of y when $x=0$.

Not always interpretable! Eg. with the heights example: height=0 is meaningless

- INTERPRETATION OF $\hat{\beta}_2$

Now consider two individuals observed at $x_1 = x_0$ and $x_2 = x_0 + 1$

The predicted values are $\hat{y}_1 = \hat{\beta}_1 + \hat{\beta}_2 x_0$

$$\hat{y}_2 = \hat{\beta}_1 + \hat{\beta}_2 (x_0 + 1)$$

so's study the difference in their predicted values

$$\hat{y}_2 - \hat{y}_1 = \hat{\beta}_1 + \hat{\beta}_2 (x_0 + 1) - \hat{\beta}_1 - \hat{\beta}_2 x_0$$

$$= \hat{\beta}_2 x_0 + \hat{\beta}_2 - \hat{\beta}_2 x_0$$

$$= \hat{\beta}_2$$

Hence $\hat{\beta}_2$ is the expected change in y when x increase of 1 unit

An equivalent interpretation holds for β_1 and β_2 in terms of the expected value of Y

we know that $E[Y_i] = \beta_1 + \beta_2 x_i$

• β_1 is the expected value of Y when $x_i = 0$: $E[Y_i | x_i = 0] = \beta_1$

• β_2 is the (additive) change in the expected value of Y when x_i is increased by 1 unit.

$$E[Y_i | x_i = x_0 + 1] - E[Y_i | x_i = x_0] = \beta_2 \quad \text{for all } x_0 \in \mathbb{R}$$

